Project Cycle 2 - Evaluation

Statistics 2 Practice | 2026 Spring

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Issues with GitHub repo

Shared folder >

1. Missing or Incorrect Folder Structure **(Major Deduction)**
   - No `report/` or `references/` folder
   - Project nested inside extra folders or zipped files
2. EDA Not Properly Saved **(Very Common)**
   - Plots shown in notebook but not saved in outputs/figures
   - Only 1 behavior plot (requirement: ≥ 2)
   - Less than 3 continuous-variable plots
3. Missing or Incorrect One-Page PDF **(High Impact)**
   - PDF not in `report/`
   - Uploaded as image(s) instead of PDF
   - Not truly “one-page visual summary”

Issues with GitHub repo

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4. Weak README **(Moderate Deduction)**
   - Missing key elements: Research question, Variables & definitions, Final conclusion, Benchmark or method explanation
5. Poor Data Organization **(Reproducibility Issue)**
   - Raw data modified or mixed with processed data
   - CSV files placed inside `notebooks/`
   - No clear distinction between raw vs cleaned data
6. Insufficient GitHub Usage **(Process Issue)**
   - Only 3 or less commits (bare minimum)
   - Large “final upload” instead of progressive work

Issues with GitHub repo

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7. Weak or Missing Documentation **(references/)**
   - No recoding rules
   - No variable definitions
   - Placeholder or empty files

- Most groups did the requested analysis, but lost marks due to:

  "Not saving / not organizing / not documenting"

- If your work cannot be followed step-by-step from GitHub:

  It is treated as incomplete, even if your statistics are correct.

Role Models

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- **Group 16** (Best example for “how to organize a reproducible project”)
  - https://github.com/hung-cc/project-cycle-2/tree/main
  - Required folders are present, notebooks are clearly separated according to the tasks, outputs are sufficient, and the PDF report is correctly placed.
- **Group 9, 15** (Best example for “active team workflow + complete submission”)
  - https://github.com/asd854026/2026-Spring-Stat2-Cycle2
  - https://github.com/Geraldand/2026SpringStat2-Cycle2/tree/main
  - Strong GitHub activity (progressive development, not one final upload).
  - Complete folders, clear README, many saved outputs, report folder, and reference files.

- I want to see **early messy work → refined analysis → final result**
- These three groups are not perfect, but they demonstrate: 1) clear workflow, 2) strong GitHub practice, 3) complete outputs, 4) reproducible structure

Issues with One-Page-Summary

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1. Too Much Text → Not a “Summary” (e.g. Group 17)
2. Too Simple → Not Enough Content (e.g. Group 10)
3. Weak Visual Hierarchy (e.g. Group 4)
4. Figures Not Supporting the Story (e.g. Group 12)
5. Results Without Interpretation (e.g. Group 11)

Good Analysis ≠ Good Summary

- A one-page summary should be:

Fast to read, clear to understand, and complete enough to justify the conclusion

Role Model #1: Group 5

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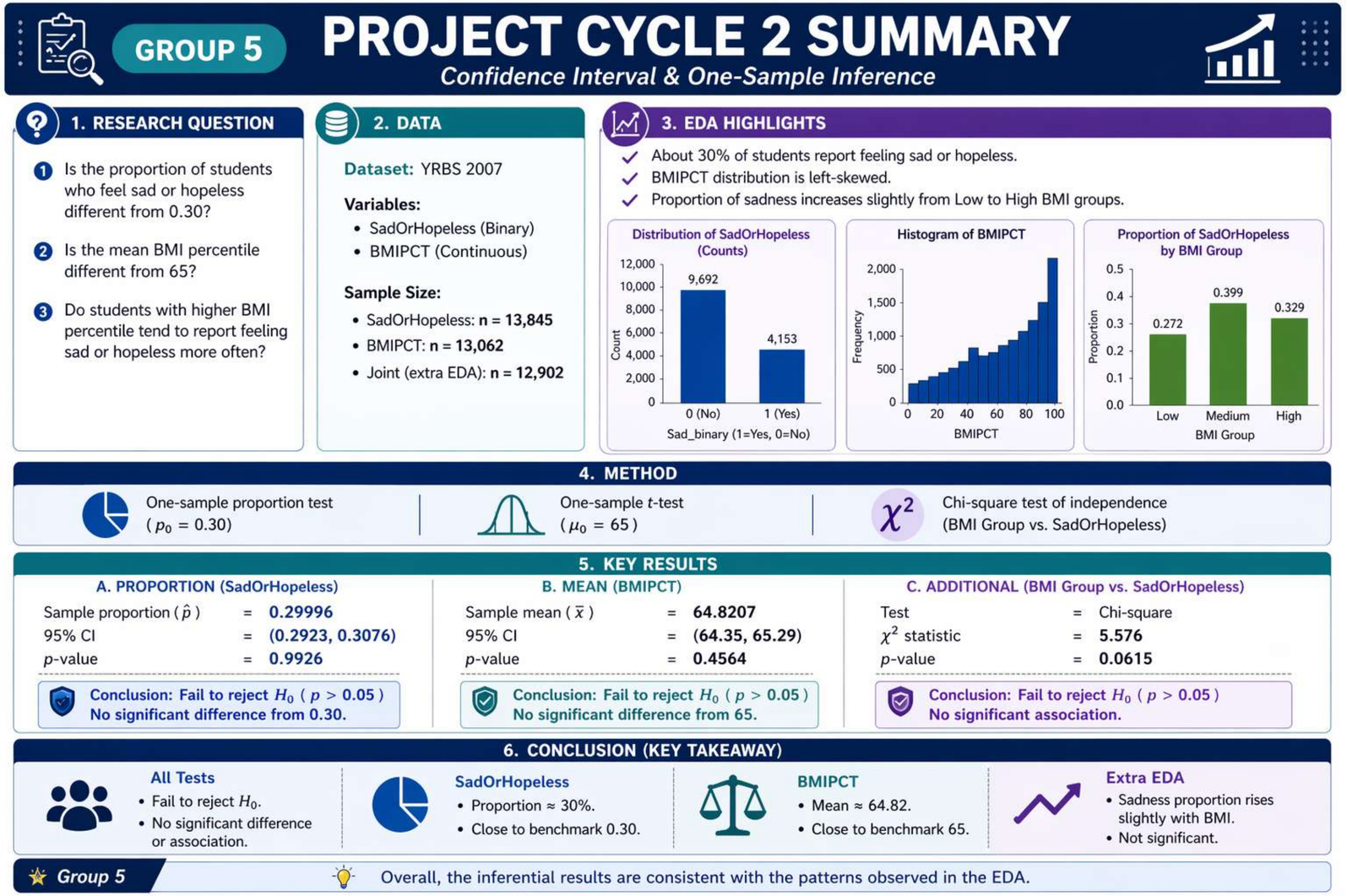
- Complete infographic: question, data, EDA, method, results, and conclusion are all clear and well organized.



Role Model #2: Group 19

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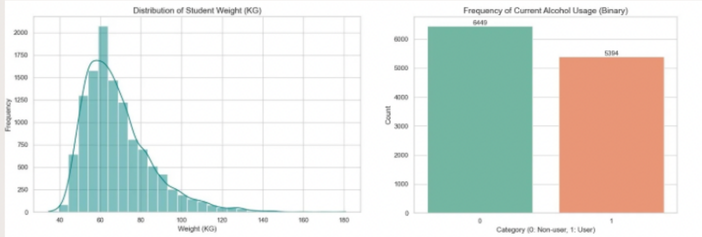
- Best balance of density, structure, inference results, EDA, and conclusion.

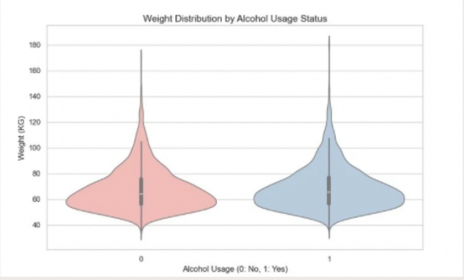
Group19 | Weight,Alcohol Use and Mental Health

1 Basic EDA & Statistical Inference



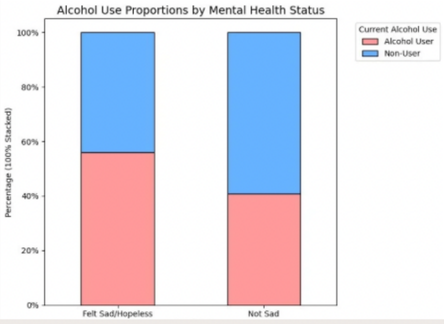
- Weight Distribution
  → Right-skewed distribution, mostly **50–70 kg**
- Alcohol Use Distribution
  → Shows overall proportion of alcohol consumption

2 Weight vs. Alcohol Use



- No clear relationship between weight and alcohol use

4 Mental Health vs. Alcohol Use



- No sadness: 40.73% drink alcohol
- Feel sad/hopeless: 55.95% drink alcohol
  → Clear increase in alcohol use among students who feel sad or hopeless

3 Statistical Inference

Alcohol Use (Proportion)
- Sample Estimate: 0.455 (45.55%)
- Benchmark: 35% p-value: < 0.001
- 95% CI: [0.446, 0.464]
- Reject H_0 (Significantly higher than 35%)

Mean Weight
- Sample Mean: 68.43 kg
- Benchmark: 68.0 kg
- p-value: 0.006
- 95% CI: [68.12, 68.73]
- Reject H_0 (Statistically significant but small difference)

Conclusion

1. **Weight shows no strong relationship with alcohol use**
2. **Mental health shows a stronger association**
3. **This supports the self-medication hypothesis, suggesting that students may use alcohol to cope with negative emotions**

Class Favorites

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- Group 15
  - 112370234 張智源
  - 113370218 彭鈺芳
  - 113370201 簡愷毅

- Group 16
  - 113370240 鍾肇紘
  - 113370206 盧軒?
  - 111370231 吳子漢